

Notes: Pancost & Robatto (RFS, 2023)

The Effects of Capital Requirements on Good and Bad Risk-Taking

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1 Big picture

- **Question.** How should bank capital requirements be set when bank deposits are held not only by households, but also by nonfinancial firms?
- **Core contribution.** The paper adds firms' precautionary demand for deposits to a quantitative general equilibrium model of capital requirements. This changes the welfare cost of reducing deposits and therefore changes the optimal capital requirement.
- **Key distinction.** The paper separates two types of risk-taking:
 - **Bad risk-taking:** banks exploit deposit insurance by expanding balance sheets and choosing higher idiosyncratic risk.
 - **Good risk-taking:** firms use safe deposits to insure against idiosyncratic cash-flow risk, which makes them more willing to hire workers and engage in risky production.
- **Main mechanism.** Raising capital requirements reduces bank deposits. In standard models, this reduction is entirely borne by households that value deposits for liquidity. In the authors' model, part of the reduction is borne by firms. The welfare cost of lower firm deposits depends on labor market adjustment.
- **Labor market insight.** If labor supply is inelastic, a fall in firms' labor demand mostly lowers wages rather than employment. Since households are both workers and owners of firms, lower wages and higher firm profits partly offset each other. Thus, the welfare cost of reducing firm deposits can be much smaller than the welfare cost of reducing household deposits.
- **Quantitative result.** In the benchmark calibration, the optimal capital requirement is 21.9% when both households and firms hold deposits, compared with 14.6% in a comparable model where households hold all deposits.
- **Policy result.** In the model with firm deposits, overshooting the true optimal requirement is not very costly. Increasing the requirement by about 10 percentage points above the optimum still preserves almost all welfare gains.
- **Robustness.** The gap between the full model and the household-only model survives many alternative calibrations: lower deposit premium, other liquid assets, different Frisch elasticities, wage rigidities, Epstein-Zin preferences, manager compensation changes, deposit preference curvature, default deadweight losses, and no risk shifting.
- **External evidence.** The paper uses the deposit channel of monetary policy to show that deposit supply reductions are associated with weaker employment and wages in high-deposit-concentration counties, consistent with the good risk-taking channel.
- **Takeaway.** Capital requirement policy should account for who holds deposits. If firms hold a substantial share, the welfare cost of reducing deposits can be lower than in household-only models, implying higher optimal capital requirements and a policy bias toward not undershooting.

2 Economic intuition: why firm deposits change capital-requirement policy

2.1 Standard household-only logic

In the standard quantitative banking models, deposits are held by households and enter household utility directly. A capital requirement increases bank equity funding and reduces deposit supply. Since households value deposits for safety and liquidity, a lower deposit supply reduces household utility.

- A deposit premium measures the value households place on deposits relative to similar safe but less liquid assets.
- A higher capital requirement lowers bad risk-taking by banks, but also reduces deposit creation.
- The optimal policy trades off lower crisis/default costs against lower liquidity services.

Interpretation: In household-only models, the welfare cost of reducing deposits is tightly linked to the observed deposit premium because all deposit value is interpreted as household liquidity utility.

2.2 Adding firms as deposit holders

The paper argues that a significant part of observed deposit demand comes from nonfinancial firms. Firms hold deposits not because deposits enter utility, but because deposits are safe assets that insure them against idiosyncratic shocks.

- Firms face uninsurable idiosyncratic productivity/cash-flow risk.
- Managers are risk averse and exposed to firm-specific risk through compensation and undiversified stakes.
- Safe deposits stabilize cash flow and make managers more willing to choose risky production, bank borrowing, capital, and labor.
- This effect is called **good risk-taking** because it increases productive activity and labor demand.

Interpretation: The same safe deposit can have different welfare value depending on who holds it. A household deposit directly produces liquidity utility; a firm deposit changes production and labor demand.

2.3 Why labor supply matters

Suppose higher capital requirements reduce the supply of deposits. If firms hold fewer deposits, their labor demand falls. The welfare effect depends on the labor supply curve:

- If labor supply is perfectly elastic, lower labor demand reduces employment, and welfare falls.
- If labor supply is perfectly inelastic, lower labor demand reduces wages but not employment.
- Since households own firms, lower wages are partly offset by higher firm profits.

Interpretation: The labor market is the general equilibrium channel. It determines whether the fall in firm deposits destroys employment or mainly redistributes income between wages and profits.

2.4 Implication for optimal capital requirements

The benefit of capital requirements is lower bank bad risk-taking. The cost is lower deposit supply. When some deposits are held by firms, the cost side is lower than in household-only models, because firm-deposit reductions are partly absorbed by wages and profits.

$$\text{Higher capital requirement} \Rightarrow \text{fewer deposits} \Rightarrow \begin{cases} \text{lower household liquidity utility,} \\ \text{lower firm good risk-taking.} \end{cases} \quad (1)$$

Interpretation: The paper's key insight is that the second channel can be less costly than the first. Hence, the regulator can set a higher capital requirement.

3 Model primitives and measurement

3.1 Agents and environment

The model is a dynamic general equilibrium economy with four types of actors:

- **Firms and managers:** firms produce using capital and labor; managers make production and portfolio choices and are exposed to firm-specific risk.
- **Banks:** banks issue deposits, hold equity, lend to firms, and may engage in risk shifting because of deposit insurance.
- **Households:** households consume, supply labor, value deposits, own firms and banks, and diversify across assets.
- **Government:** the government provides deposit insurance and taxes households to repay depositors of failed banks.

Interpretation: The model is designed so that deposits are simultaneously household liquidity assets, firm precautionary assets, and bank liabilities that create moral hazard.

3.2 Firm balance sheet and deposit holdings

Each firm i begins the period with wealth a_t^i , which the manager allocates between safe deposits d_t^i and directly held productive capital k_t^i :

$$d_t^i + k_t^i \leq a_t^i. \quad (1)$$

Deposits pay a safe gross return R_t^d because they are insured by the government. The firm also borrows b_t^i from banks, subject to

$$b_t^i \leq \xi k_t^i, \quad 0 < \xi < 1. \quad (2)$$

Interpretation: The borrowing constraint links bank credit to the firm's own capital. Because bank loans are partly funded with liquidity-premium deposits, firms want to borrow up to the constraint.

3.3 Firm production and cash flow

Production is

$$y_t^i = A_{t+1} z_{t+1}^i (k_t^i + b_t^i)^\gamma (l_t^i)^{1-\gamma}, \quad (3)$$

where A_{t+1} is aggregate productivity, z_{t+1}^i is idiosyncratic productivity, $k_t^i + b_t^i$ is total productive capital, and l_t^i is labor.

At the start of $t + 1$, firm wealth is

$$x_{t+1}^i = y_t^i - w_t l_t^i - r_t b_t^i + R_t^d a_t^i + (1 - \delta) z_{t+1}^i k_t^i. \quad (4)$$

Interpretation: Deposits are the only safe asset on the firm side. They reduce the risk that low idiosyncratic productivity prevents the firm from paying workers and creditors.

3.4 Dividend policy and manager compensation

Shareholders choose the payout fraction α_{t+1}^i . Wealth retained inside the firm is subject to dividend adjustment costs:

$$a_{t+1}^i = (1 - \alpha_{t+1}^i) x_{t+1}^i - \frac{\varphi}{2} x_{t+1}^i (\alpha_{t+1}^i - \bar{\alpha})^2. \quad (6)$$

Manager compensation is

$$c_{t+1}^i = \kappa [\theta_f a_t^i + \theta_e \alpha_{t+1}^i x_{t+1}^i + \theta_o \max\{0, a_{t+1}^i - a_t^i\}]. \quad (7)$$

Interpretation: The manager is imperfectly diversified because her consumption depends on the firm. This agency friction generates precautionary demand for safe deposits at the firm level.

3.5 Manager's problem and good risk-taking

The manager chooses deposits, capital, labor, borrowing, compensation, and next-period assets to maximize

$$V_t^m(a_t^i) = \max E_t \left[\beta^m \frac{(c_{t+1}^i)^{1-\gamma_m}}{1-\gamma_m} + V_{t+1}^m(a_{t+1}^i) \right]. \quad (8)$$

Proposition 1 shows that when the borrowing constraint binds, optimal choices satisfy

$$k_t^i = \phi_t^k a_t^i, \quad l_t^i = \phi_t^l a_t^i, \quad b_t^i = \xi k_t^i. \quad (9-12)$$

Interpretation: The firm side aggregates because the optimal capital and labor ratios are independent of firm size. Deposits matter through the manager's risk-bearing: safer cash flow encourages more productive risk-taking.

3.6 Banks and bad risk-taking

Banks live for one period. A bank receives net worth n_t , issues deposits d_t , and buys bank assets b_t :

$$b_t = d_t + n_t. \quad (16)$$

The capital requirement is

$$n_t \geq \zeta \frac{b_t}{1 - \nu p_t}, \quad (17)$$

where ζ is the capital requirement and p_t is the risk-shifting choice.

Bank profits are affected by an idiosyncratic shock ε and by risk-shifting costs $\lambda(p_t)$:

$$\max_{b_t, d_t, p_t \geq 0} E_t \int_{\varepsilon_t}^{\infty} [\varepsilon b_t (1 - \delta + r_t - \lambda(p_t) r_t) - R_t^d d_t] dF(\varepsilon; p_t). \quad (15)$$

Interpretation: Deposit insurance and limited liability make risk shifting privately attractive. Capital requirements reduce the incentive to exploit insured deposits.

3.7 Households

Households consume, work, hold deposits, bank equity, and firm equity. They receive utility from deposits as in the capital-requirement literature. The deposit premium is pinned down by households' marginal utility of liquidity services and by firms' precautionary demand.

Interpretation: The key calibration issue is that the observed deposit premium is not entirely household utility. A part of it reflects firm demand for safe assets.

3.8 Labor market and wage rigidity

The actual wage is a weighted average of the previous wage and the frictionless wage:

$$w_t = \omega w_{t-1} + (1 - \omega) w_t^f. \quad (27)$$

The frictionless wage satisfies a household labor-supply condition:

$$w_t^f E_t \{\Lambda_{t+1}\} = \chi l_t^{1/\eta}. \quad (28)$$

Interpretation: Wage rigidities make the analysis conservative because they prevent wages from fully absorbing the reduction in firm labor demand after tighter capital requirements.

3.9 Government deposit insurance

The government taxes households to repay insured depositors at failed banks. Taxes are

$$T_{t+1} = \int_{-\infty}^{\varepsilon_t} [R_t^d d_t - \varepsilon b_t (1 - \delta + r_t - \lambda(p_t) r_t)] dF(\varepsilon; p_t). \quad (29)$$

Interpretation: Deposit insurance creates bad risk-taking, but it also makes deposits safe, which creates the household liquidity service and the firm insurance role.

3.10 Aggregate risk and resource constraint

Aggregate productivity follows

$$\log A_t = (1 - \rho) \log \bar{A} + \rho \log A_{t-1} + \sigma_A \varepsilon_t^A. \quad (2)$$

The aggregate resource constraint is

$$\begin{aligned} c_t + (k_t + b_t) &\leq (1 - \delta)(k_{t-1} + b_{t-1}) + A_t(k_{t-1} + b_{t-1})^\gamma l_{t-1}^{1-\gamma} \\ &\quad - \lambda(p_{t-1}) r_{t-1} b_{t-1} - \frac{\varphi}{2} x_t (\alpha_t - \bar{\alpha})^2. \end{aligned} \quad (3)$$

Interpretation: Resources are used for consumption and capital accumulation, net of the real costs of bank risk shifting and dividend adjustment.

3.11 Welfare

Welfare is formally

$$W = V_0^h(a_0^h) + \theta \int V_0^m(a_0^i) di. \quad (30)$$

The authors focus on the limiting case in which managers' welfare and consumption are negligible relative to households, so that optimal capital requirements are computed from household welfare.

Interpretation: This keeps the welfare analysis transparent: the objective is the representative household's welfare, while manager incentives still shape firm behavior.

4 Calibration and quantitative analysis

4.1 Fixed parameters

The paper fixes a first group of parameters using standard values and independent evidence:

- Discount factors: $\beta = \beta^m = 0.95$.
- Capital share: $\gamma = 0.3$.
- Depreciation: $\delta = 0.1$.
- Productivity persistence: $\rho = 0.95$.
- Household and manager risk aversion: $\gamma_c = \gamma_m = 3$.
- Deposit preference curvature: $\gamma_d = 1.4$.
- Frisch elasticity: $\eta = 1$.
- Manager compensation weights: $\theta_e = 1$, $\theta_f = 0.0297$, $\theta_o = 0.0960$.

Interpretation: The model is disciplined by both macro-finance parameters and micro evidence on CEO compensation, because manager incentives are central to the firm's precautionary demand for deposits.

4.2 Parameters matched to moments

The calibrated parameters include household deposit preference ψ , pledgeability ξ , bank idiosyncratic risk σ , risk-shifting cost coefficients λ_1, λ_2 , wage rigidity ω , risk-weight sensitivity ν , aggregate volatility σ_A , firm idiosyncratic shock parameters z_L, p_z , and dividend adjustment cost φ .

Target moments include:

- deposit premium of 1.12%;
- deposits-to-GDP ratio of 0.43;
- average bank default rate of 0.61%;
- 68% low-default times;
- 85th percentile bank default rate of 1.55%;
- volatility of employment relative to GDP of 0.58;
- volatility of risk-weighted assets relative to assets of 1.27;

- volatility of log GDP of 1.75%;
- household share of deposits of 66.4%;
- continuers' employment growth of 2.50%;
- firm-deposit volatility relative to GDP of 3.49.

Interpretation: The restricted “HH only” calibration is forced to match the same deposit premium and aggregate deposit ratio, but it assigns all deposits to households.

4.3 Bank risk-shifting functional forms

The cost of bank risk shifting is approximated by

$$\lambda(p_t) = \lambda_0 + \lambda_1 p_t + \lambda_2 p_t^2, \quad (4)$$

with $\lambda_0 = 0$. The idiosyncratic bank shock is a mixture: a default-like shock with probability p_t and a lognormal shock with probability $1 - p_t$.

Interpretation: Risk shifting produces normal times and crisis times: when $p_t = 0$, defaults are low; when $p_t > 0$, default rates spike.

4.4 Untargeted moments

The authors compare the full and household-only models to untargeted moments such as the volatility of investment, consumption, wages, firm deposits, household deposits, bank assets, and default rates; correlations with GDP; correlations between deposit premiums and deposits/labor; and FDIC resolution costs.

Interpretation: The full model performs similarly to the restricted model on broad macro moments but adds realistic behavior for firm deposits and labor because firms directly demand safe assets.

4.5 Welfare experiment

The economy begins at an 8% capital requirement. The authors then change ζ and compute transition welfare to the new stochastic steady state. Welfare is measured in consumption-equivalent units.

- Full model optimum: 21.9%.
- Household-only optimum: 14.6%.
- Welfare gain at full-model optimum: 0.224% of annual consumption.
- Welfare gain at household-only optimum: 0.028% of annual consumption.

Interpretation: Including firm deposits raises the optimal capital requirement and changes the shape of the welfare function.

4.6 Overshooting versus undershooting

In the household-only model, setting the requirement far above the optimum erases the gains. In the full model, the welfare curve is almost flat above the optimum.

Interpretation: If a regulator is uncertain about the true optimum, the model suggests that overshooting is less costly than undershooting, because higher requirements continue to restrain bank risk-taking while firm-deposit costs are muted by labor-market adjustment.

5 External empirical evidence

5.1 Deposit channel of monetary policy

The paper builds on Drechsler, Savov, and Schnabl (2017), who show that increases in the federal funds rate reduce deposits more in counties with concentrated banking markets. Bank market power allows banks to widen deposit spreads, causing deposits to flow out.

Interpretation: The deposit channel provides exogenous-looking cross-county variation in deposit supply, which helps test whether deposit reductions affect employment and wages.

5.2 County-level regression

The baseline regression is

$$y_{c,t} = \alpha_c + \alpha_t + \beta \Delta FF_t \times HHI_c + \varepsilon_{c,t}. \quad (33)$$

where $y_{c,t}$ is either employment growth or wage growth, ΔFF_t is the change in the Fed funds rate, and HHI_c measures county-level deposit concentration.

Interpretation: A negative coefficient means that monetary tightening reduces employment or wages more in counties where deposits are more exposed to the deposit channel.

5.3 Controlling for lending

The paper also includes a lending proxy from DSS to separate the direct effect of deposit shortages on firm good risk-taking from the conventional bank lending channel.

Interpretation: The authors want to show that deposits matter not only because banks lend less, but also because firms' own safe asset holdings affect labor demand.

6 Tables and figures

Visuals are directly extracted from the paper and grouped compactly.

6.1 Figure 1: labor-market intuition

Read:

- The figure contrasts two extreme labor supply cases.
- With perfectly elastic labor supply, a fall in labor demand reduces employment one-for-one.
- With perfectly inelastic labor supply, a fall in labor demand mostly reduces wages.
- If households own firms, lower wages are partly offset by higher firm profits.

Economic message: The welfare cost of reducing firm deposits depends on labor-market adjustment. This is the key general equilibrium channel that raises the optimal capital requirement.

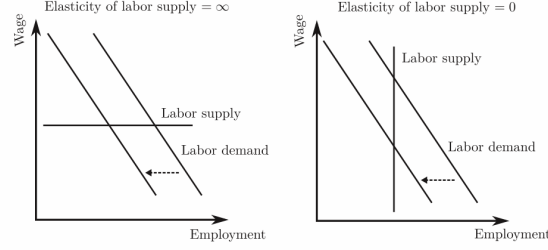


Figure 1
Equilibrium in the labor market and effects of a change in labor demand
Left panel, infinitely elastic labor supply; right panel, fully inelastic labor supply.

6.2 Table 1 and Figure 2: calibration and bank defaults

Read:

- Table 1 reports fixed parameters and matched-moment parameters for the full and household-only calibrations.
- In the full calibration, households hold 66.4% of deposits and firms hold the remaining share.
- The model matches the 1.12% deposit premium, deposits-to-GDP ratio of 0.43, average bank default rate of 0.61%, and several volatility moments.
- Figure 2 compares sorted bank default rates in the data and the model.
- The model matches the large default-rate spikes because banks can engage in risk shifting.

Economic message: The calibration is designed to match both the deposit side of the economy and the crisis/default behavior of banks.

Table 1
Calibrated parameter values

A. Preset parameters

Parameter	Value	Parameter	Value	Parameter	Value
η	1	γ	0.3	$\bar{A}$	1
β, β^m	0.95	γ_c	3	$E\{z^c\}$	1
δ	0.1	ρ	0.95	χ	1
$\bar{\alpha}$	0.05	γ_m	3	γ_d	1.4
θ_e	1	θ_f	0.0297	θ_o	0.0960

B. Parameters chosen to match data moments

Parameter	Value	Moment target	Moment value
	Full	HH only	Data & full
ψ	0.00248	0.00446	Deposit premium, $R^f - R^d$
ξ	0.3002	0.3054	Deposits-to-GDP ratio
σ	0.027	0.0275	Average bank default rate
λ_1	7.29281	7.42856	Low-default times
λ_2	2.68	3.22	p85 bank default rate
ω	0.623	0.619	Vol(employment)/vol(GDP)
ν	0.80	0.99	Vol(risk-weighted assets)/vol(assets)
σ_A	0.0142	0.01465	Volatility logGDP
z^L	0.90235	1	Fraction of households' deposits
p_z	0.838	1	Continuers employment growth
φ	1.32	0	Vol(firm deposits)/vol(GDP)

In panel B, "Full" denotes the full calibration with both households' and firms' deposits, and "HH only" denotes the restricted calibration in which all the deposits are held by households. All volatilities are computed by taking the logarithm and applying the HP-filter with parameter 100, both in the model and in the data.

Table 1: calibrated parameters

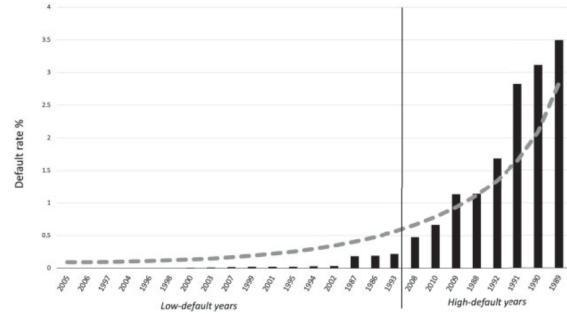


Figure 2
Bank default rates, sorted, in the data (bars) and model (dashed line)

Banks' default rate in the data is measured as the fraction of banks that are taken over by the FDIC (weighted by assets) and for which the FDIC incurred a positive resolution cost. In the model, we sort the default rate in each simulated path and then average it out across paths.

Figure 2: bank default rates

6.3 Table 2 and Figure 3: untargeted moments and welfare

Read:

- Table 2 reports untargeted business-cycle moments and correlations.
- The model performs well on volatility of investment, wages, bank default rates, and deposit-related correlations.
- Figure 3 shows welfare as a function of the capital requirement.

- The full model optimum is 21.9%, while the household-only optimum is 14.6%.
- The full-model welfare curve is much flatter above the optimum.

Economic message: The core quantitative result is visible in Figure 3: firm deposits raise the optimal capital requirement and make overshooting much less costly.

Table 2

Untargeted moments

Moment (untargeted)	Data	Full	HH only
Volatilities, ratios to log GDP, HP-filtered			
log(investment)	3.77	3.61	3.63
log(consumption)	0.69	0.41	0.41
log(wages)	0.77	0.76	0.77
log(firm deposits / labor)	3.27	2.94	—
Bank profits / GDP	0.17	0.18	0.14
log(1+deposit premium)	0.52	0.04	0.01
Correlations with log GDP, HP-filtered			
log(investment)	0.92	0.98	0.98
log(consumption)	0.90	0.89	0.89
log(wages)	0.40	0.86	0.87
log(firm deposits)	0.67	0.35	—
log(labor)	0.89	0.20	0.22
log(firm deposits / labor)	0.55	0.38	—
Bank profits / GDP	0.53	0.34	0.35
log(lagged GDP)	0.68	0.47	0.47
Other correlations, HP-filtered			
Corr(log[1+deposit premium], log(deposits))	-0.22	-0.30	0.21
Corr(log[1+deposit premium], log(labor))	0.82	0.95	0.70
Corr(log[1+deposit premium], log(wages))	-0.26	0.23	0.69
Corr($\Delta \log(\text{deposits})$, $\log(\text{labor})^d$)	0.76	0.33	0.33
Corr($\Delta \log(\text{deposits})$, $\log(\text{wages})^d$)	0.33	0.58	0.59
Corr(log[risk-weighted assets], bank default rate $t+1^d$)	0.44	0.53	0.45
Corr(log[risk-weighted/total assets], bank default rate $t+1^d$)	0.39	0.84	0.75
Other moments, not HP-filtered			
Vol. household deposits / GDP, every 3 years	2.98	2.10	0.98
Vol. bank default rate	1.06	0.76	0.64
FDIC resolution costs to assets of failed banks	19%	28%	27%
Corr(bank default rate, $\Delta \log \text{GDP}$)	-0.27	-0.30	-0.25
Banks' assets, % of capital stock	24.4	23.1	23.4

The moments in the first three panels are computed by applying the HP-filter both to the data and model-simulated series, using a smoothing parameter of 100. d : $\Delta \log$ deposits and bank default rate are not HP-filtered. "Full" denotes the full calibration with both households' and firms' deposits, and "HH only" denotes the restricted calibration in which households hold all the deposits.

Table 2: untargeted moments

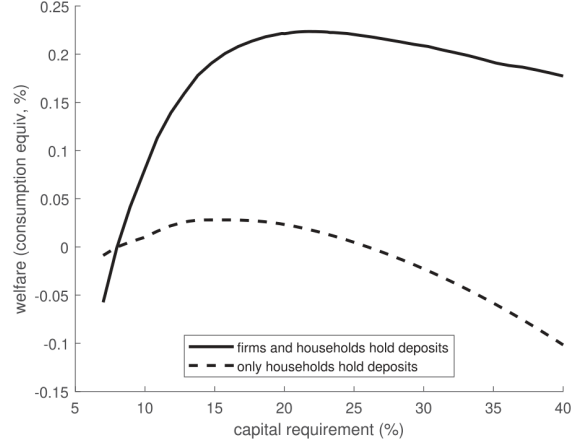


Figure 3
Welfare results

The solid line plots welfare $W(\zeta)$ for the full model calibration in which households and firms hold deposits, and the dotted line plots the welfare for the restricted calibration in which households hold all the deposits. Parameter values are in Table 1. Welfare is defined in consumption-equivalent units. We compute welfare by averaging across 3,000 simulated economies for 300 periods each, where each of the 3,000 simulations is initialized by drawing a point in the state space of the ergodic distribution of the $\zeta=8\%$ economy.

Figure 3: welfare results

6.4 Table 3 and Table 4: steady-state effects and robustness

Read:

- Table 3 shows that raising ζ from 8% to 21.9% sharply reduces bank defaults, especially defaults due to risk shifting.
- Deposits fall, but bank assets and macro quantities move only modestly.
- Firms' deposits fall more than households' deposits because firm deposit demand is more price elastic.
- Table 4 shows that the higher optimal requirement in the full model is robust across alternative assumptions.
- The gap between full and household-only optima is especially sensitive to the Frisch elasticity and wage rigidities.

Economic message: Capital requirements do what they are supposed to do on the bank side, but the real-side cost is limited when firm-deposit reductions are absorbed by wages rather than employment.

Table 3
Capital requirements and steady-state values

	Capital requirement			
	8%	15%	21.9%	25%
Bank default (% of all banks)	0.61	0.11	0.02	0.01
Bank default due to risk shifting (% of all banks)	0.53	0.11	0.02	0.01
Bank rental rate of capital, net of depreciation ($r_f - \delta$, %)	4.20	4.24	4.28	4.30
Deposit premium (%)	1.12	1.18	1.23	1.25
Change from $\zeta = 8\%$, percent				
Total deposits	-7.8	-15.3	-18.7	
Households' deposits	-3.79	-6.8	-7.93	
Firms' deposits	-15.9	-32.3	-40.2	
Bank equity	+85.6	+170	+208	
Bank assets	-0.27	-0.29	-0.28	
Output	-0.02	-0.02	-0.01	
Labor	-0.20	-0.25	-0.26	
Capital	-0.27	-0.29	-0.28	
Capital-labor ratio	-0.06	-0.03	-0.003	
Consumption	0.04	0.06	0.05	
Wages	-0.03	-0.04	-0.04	
Return on firm assets R_{t+1}^f , Equation (14)	0.02	0.02	0.02	
Labor share of income	-0.22	-0.29	-0.30	
Capital and profit share of income ($= 1 - \text{labor share}$)	0.51	0.65	0.69	

The table reports average values from the stochastic steady-state distribution, with the value of the capital requirement ζ indicated in the top row. The top panel reports values as percentages, and the bottom panel reports values in percent deviations from the $\zeta = 8\%$ economy.

Table 3: capital requirements and steady states

Table 4
Robustness exercises

Calibration	Optimal capital requirement		
	Full (%)	HH only (%)	Difference
Benchmark	21.9	14.6	7.3
Deposit premium = 0.56%	30.8	27.9	2.9
Other liquid assets held by firms = 1.64% of GDP	22.5	14.6	7.9
Frisch elasticity $\eta = 1.5$	21.0	17.2	3.8
Frisch elasticity $\eta = 0.5$	24.4	14.2	10.2
More wage frictions, $\text{vol}(w)/\text{vol}(\text{GDP}) = 0.73$	21.6	19.6	2.0
Less wage frictions, $\text{vol}(w)/\text{vol}(\text{GDP}) = 0.80$	24.6	14.8	9.8
Epstein-Zin preferences, $\text{EIS} = 1.5$	22.5	14.6	7.9
Lower manager risk aversion, $\gamma_m = 2$	22.6	14.6	8.0
Only equity compensation, $\theta_f = \theta_d = 0$	22.1	14.6	7.5
Higher fixed and option compensation, $2 \times$	21.9	14.6	7.3
Households' preferences for deposits, $\gamma_d = 0.8$	26.4	20.5	5.9
Deadweight loss from default $\sigma = 4.5\%$	24.5	15.7	8.8
Share of households' deposits 82.8%	24.5	14.6	9.9
No risk shifting, $\lambda_t = \infty$	3.2	1.9	1.3

The column "Full" indicates the optimal capital requirement in the full calibration with both households' and firms' deposits. The column "HH only" indicates the optimal capital requirement in the restricted calibration in which households hold all the deposits.

Table 4: robustness exercises

6.5 Table 5: independent evidence for good risk-taking

Read:

- The data are county-year observations from 1997 to 2013.
- The main interaction is $\Delta FF_t \times HHI_c$.
- Employment growth falls more after monetary tightening in high-deposit-concentration counties.
- Wages also fall more in those counties.
- The results survive controls for the lending proxy, county fixed effects, time fixed effects, and county-by-ZLB fixed effects.

Economic message: This evidence supports the model's good risk-taking channel: when deposit supply falls, firms reduce labor demand, affecting both employment and wages.

Table 5
Deposit channel and good risk-taking

	(1) $\Delta I_{c,t}$	(2) $\Delta I_{c,t}$	(3) $\Delta I_{c,t}$	(4) $\Delta w_{c,t}$	(5) $\Delta w_{c,t}$	(6) $\Delta w_{c,t}$
$\Delta FF_t \times HHI_c$	-0.005*** [0.001]	-0.005*** [0.001]	-0.005*** [0.001]	-0.701** [0.352]	-0.870** [0.349]	-0.673* [0.365]
$\Delta FF_t \times (\text{Lending proxy})_{c,t-1}$			-0.008** [0.003]			-2.170** [1.051]
$(\text{Lending proxy})_{c,t-1}$			0.003 [0.007]			-0.870 [2.862]
Time FE	Y	Y	Y	Y	Y	Y
County FE	Y	Y	Y	Y	Y	Y
County $\times$ ZLB FE	N	Y	Y	N	Y	Y
Observations	54,103	54,103	54,103	54,103	54,103	54,103
R-squared	.147	.202	.202	.108	.163	.163

The data are at the county-year level for the period 1997 to 2013. $\Delta I_{c,t}$ is the logarithmic change in total employment as in DSS. $\Delta w_{c,t}$ is the change in the annual average weekly wage in county c , time t . ΔFF_t is the change in Fed funds rate. HHI_c is the deposit Herfindahl index in county c as in DSS. $(\text{Lending proxy})_{c,t-1}$ corresponds to the variable "County-HHI" in DSS and is constructed as follows: for each bank b , let Bank-HHI_{t-1}^b be the average HHI_c of the counties in which bank b operates, weighted by deposits; then, $(\text{Lending proxy})_{c,t-1}$ is the average Bank-HHI_{t-1}^b for all the banks that operate in county c , weighted by deposits. Standard errors are clustered by county. * $p < .1$; ** $p < .05$; *** $p < .01$.

7 Short summary

- The paper studies optimal capital requirements when both households and firms hold bank deposits.
- Deposits create household liquidity utility and also provide firms with safe assets that insure against idiosyncratic risk.
- Firm deposits support good risk-taking: firms hire more and produce more when deposits reduce cash-flow volatility.

- Bank deposit insurance induces bad risk-taking: banks expand balance sheets and increase risk shifting.
- Capital requirements reduce bad risk-taking but reduce deposit supply.
- In a household-only model, reducing deposits is costly because all deposit value is household liquidity utility.
- In the full model, part of the deposit reduction is borne by firms, and the welfare cost depends on labor-market adjustment.
- With inelastic labor supply, reduced firm labor demand mostly lowers wages, and the wage loss is partly offset by higher firm profits.
- The benchmark optimal capital requirement is 21.9% in the full model and 14.6% in the household-only model.
- Overshooting the full-model optimum is much less costly than in the household-only model.
- Robustness checks support the importance of labor supply elasticity and wage rigidity.
- County-level evidence using the deposit channel of monetary policy supports the good risk-taking mechanism.

8 Conclusion

8.1 Core conclusion

Pancost and Robatto show that optimal bank capital requirements are higher when nonfinancial firms hold deposits. Firm deposits support productive risk-taking by insuring firms against idiosyncratic shocks. Since the welfare cost of reducing firm deposits is partly absorbed through wages and profits, the cost of tighter capital requirements is lower than in household-only models.

8.2 Economic intuition

Capital requirements reduce banks' bad risk-taking by limiting balance-sheet expansion and risk shifting. They also reduce deposits. If deposits are held only by households, this reduction directly lowers household liquidity utility. If some deposits are held by firms, lower deposits reduce firms' labor demand. But with relatively inelastic labor supply, wages adjust downward and employment falls only modestly. Because households own firms, the wage loss is partly offset by firm profits.

8.3 Policy takeaway

The paper argues that models ignoring firm deposits may understate the optimal capital requirement and overstate the cost of setting requirements above the optimum. For policymakers, the result suggests that uncertainty about the true optimum should not automatically lead to caution against higher capital requirements; in this model, undershooting is more costly than overshooting.